

Shortest Paths with Negative Cycles

Topic.

Shortest paths, Bellman–Ford algorithm, negative cycle detection.

Statement.

Given a weighted directed graph and a starting vertex s .

For every vertex u , find the length of the shortest path from s to u .

The graph may contain multiple edges and loops. Edge weights can be negative.

For each vertex u , output:

- the length of the shortest path from s to u ;
- $*$, if there is no path from s to u ;
- $-$, if the shortest path from s to u does not exist because the value can be made arbitrarily small.

Input.

The first line contains three integers

$$n, m, s,$$

where n is the number of vertices, m is the number of edges, and s is the starting vertex.

Each of the next m lines contains three values:

$$a, b, w,$$

meaning that there is a directed edge from a to b with weight w .

Output.

Print n lines.

For each vertex u , print the corresponding answer:

- the shortest distance from s to u ;
- $*$, if u is unreachable from s ;
- $-$, if there is no finite shortest path to u .

Implementation.

The implementation is provided separately in the file

`solution.cpp`.